

Bradley Heim

Glen Ellyn, IL 60137 | 331-551-1156 | heimb@purdue.edu | bradleyheim.github.io/portfolio/

EDUCATION

Purdue University

West Lafayette, IN

B.S. in Aeronautical & Astronautical Engineering, College of Engineering

May 2027

B.S. in Integrated Business and Engineering, Mitch Daniels School of Business

3.73/4.0 GPA

Concentrations: **Data Analytics** and **Leadership via Larsen Leaders Academy**

Relevant Coursework: Fluid Dynamics, Thermodynamics, Structural Analysis, Mechanics of Materials, Partial Differential Equations, Graphical Communication & Spatial Analysis, Signals Analysis, Introduction to Aerospace Design, Probability & Statistics in Engineering

Study Abroad: Barcelona, Spain (Summer 2025) - Collaborated with the CEO of Spanish startup Two.Zero to develop a growth strategy, applied organizational behavior and cross-cultural management principles to deliver actionable business recommendations

TECHNICAL EXPERIENCE

Medical Instrumentation in Space

West Lafayette, IN

Project Lead

August 2025 – Present

- Led a 7-person multidisciplinary team designing a peristaltic infusion pump for blood transfusion in microgravity, integrating fluid mechanics, mechanical design, and electronics for safe spaceflight operation, producing weekly technical documentation
- Authored a technical proposal for the NASA Human Landing Challenge, defining system requirements and communicating engineering rationale to external reviewers while managing project timeline and overseeing initial pump validation testing

Purdue Space Program - Satellites

West Lafayette, IN

Structures, Mechanisms and Thermals

August 2024 – Present

- Support structures and thermals on a 3U CubeSat built to California Polytechnic standards, contributing to machine shop manufacturing including proposing and implementing a vacuum chuck fixture for CNC machining of satellite assemblies
- Advancing toward Preliminary Design Review following successful System Design Review, gaining exposure to industry thermal analysis tools including ANSYS and Thermal Desktop

AAE 251 Aircraft Design Competition

West Lafayette, IN

1st place team

January - May 2025

- Designed an unmanned tanker aircraft for wildfire retardant delivery, applying systems engineering practices including risk analysis performing aerodynamic sizing, propulsion selection, and mission performance analysis in MATLAB, co-authoring a 58-page technical design report and recorded video presentation for competition judges
- Awarded 1st Place in aircraft design across all classes, nominated by Purdue faculty and selected by Northrop Grumman alumni

Design For Good Hackathon sponsored by Dormakaba

West Lafayette, IN

1st place team

October 2024

- Won multidisciplinary innovation design competition with 120 IBE students and 20 Industrial Design students across 14 teams
- Undertook industrial design process to ideate, prototype and present to 7 judges an access solution for individuals with disabilities

Purdue Lunabotics

West Lafayette, IN

Systems Team

August 2023 – May 2024

- Developed systems engineering documentation and requirements templates to standardize deliverables across subteams, ensuring compliance with NASA Lunabotics competition requirements and internal team standards
- Co-authored a systems engineering paper placing 3rd among 58 collegiate teams, and placed 2nd in presentation and design

LEADERSHIP EXPERIENCE

Zeta Beta Tau Alpha Chapter

West Lafayette, IN

Housing Manager

July 2025 – Present

- Oversee facility management for 80 residents, with an annual budget of \$8000 ensuring proper maintenance and regular cleanings
- Planned and executed 7 member-led projects across 60 participants, managing a \$1,500 budget and delegating responsibilities to deliver results efficiently and within financial constraints

Operations Director

November 2024 – June 2025

- Directed operations for a 180-member chapter, ensuring full compliance with Purdue FSCL and ZBT International requirements while mentoring chairmen in academics, judicial processes, and chapter performance to improve efficiency and chapter GPA
- Designed and implemented a comprehensive database system to efficiently track and analyze member participation and standards across 12 parameters, enhancing data accessibility, accountability, and executive decision-making

JOB EXPERIENCE

Purdue Football Student Video Manager

March 2024 – Present

CU1 Amphitheatre Concert Security - Roamer

June 2024 – August 2025

RELEVANT SKILLS

NX (80 hours), Fusion, Python, Matlab, Java, R, ANSYS, Microsoft Office, Aras Innovator, Jira, Confluence, Technical Writing, Project Management